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Warming Asymmetry in Climate Change Simulations

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Abstract. Climate change simulations made with coupled global climate models typically show a marked hemispheric asymmetry with more warming in the northern high latitudes than in the south. This asymmetry is ascribed to heat uptake by the ocean at high southern latitudes. A recent version of the CCCma climate model exhibits a much more symmetric warming, compared to an earlier version, and agrees somewhat better with observed 20th century trends. This is associated with an improved parameterization of ocean mixing which results in a decrease in heat penetration into the Southern Ocean, in accord with earlier ocean-only and simple coupled model investigations. The global average warming and the net penetration of heat into the global ocean (and hence its thermal expansion) are essentially unchanged. Observed trends in sea-ice extent over the past two decades exhibit hemispheric asymmetry with a statistically significant decrease in northern but not in southern ice cover. Both model versions are consistent with these observations implying that observed ice extent is not yet an indicator of asymmetry in future global warming. Taken together, these results suggest that southern hemisphere climate warming at a rate comparable to that in the northern hemisphere should be considered a realistic possiblity.

1. Introduction

Figure 1a displays the temperature change from 1971-1990 to 2041-2060 simulated with CGCM1, the first version of the Canadian Centre for Climate Modelling and Analysis (CCCma) coupled climate model, forced with historical and projected greenhouse gas (GHG) and aerosol forcing (*Boer et al.*,[2000a, b], hereafter referred to as BFR). The asymmetry in temperature response is typical of the current generation of coupled models (e.g. *Kattenberg et al.*,[1996]) and is mainly due to deeper vertical mixing in the Southern Ocean which sequesters heat in the ocean around Antarctica.

Ocean-only model results, like those of *Robitaille and Weaver* [1995] and *McDougall et al.* [1996], indicate that parameterizing sub-grid-scale ocean mixing as horizontal /vertical (HV) diffusion results in excessive mid-depth heat transport to high latitudes and leads to excessive deep convection in the Southern Ocean. Such experiments also illustrate that the more physically based isospycnal, eddy-stirring mixing parameterization of *Gent and McWilliams* [1990] (GM) reduces this convective mixing and leads to more realistic ventilation of the high-latitude oceans. Results like these have led to speculation that use of the GM scheme in coupled climate models would reduce the heat sequestered in the Southern Ocean and result in greater surface warming, a more symmetrical warming pattern, and an increase in global mean warming.

Here we compare results obtained with two versions of the CCCma global coupled model. The same atmospheric component is used in both cases. It is a spectral model with T32 truncation and 10 unequally-spaced vertical levels (*McFarlane et al.*, 1992), forced with the 1900 to 2100 GHG concentrations and aerosol loadings of *Mitchell et al.*, 1995, as described in BFR. The ocean component in both cases is a grid-point model with 1.875°resolution and 29 vertical levels, based on the GFDL MOM1.1 code (*Pacanowski et al.*, 1993). The first version, CGCM1, and its control climate are described in some detail by *Flato et al.* [2000]. The horizontal diffusivity in CGCM1 is $2 \times 10^3 m^2/s$ and the vertical diffusivity is $3 \times 10^{-5} m^2/s$. The second version, CGCM2, differs from CGCM1 by using GM rather than HV ocean mixing and has an along-isopycnal diffusivity of $1 \times 10^3 m^2/s$, a background horizontal diffusivity of 1/10 the isopycnal value, and a vertical diffusivity of $5 \times 10^{-5} m^2/s$. CGCM2 uses the cavitating fluid sea-ice dynamics scheme of *Flato and Hibler* [1992] in place of the thermodynamic-only sea ice in CGCM1, and there are are some technical differences in the ocean spin-up procedure and in the manner in which the monthly-mean heat and moisture flux adjustments are computed.

2. Results

The change in surface air temperature at the middle of the 21st century simulated by the two model versions is shown in Figure 1a and b. Both show greater warming over land than over ocean and slight cooling in the northwest Atlantic (a result associated with changing ocean convection patterns). The most striking difference is the substantial Southern Ocean warming produced by CGCM2, compared to weak warming and patches of cooling produced by CGCM1. This is especially evident in the zonal average temperature change displayed in Figure 1c. CGCM2 was developed as an improved model version for the purpose of climate change simulations rather than as a sensitivity test of ocean mixing parameterization as such, however its results may be compared with more controlled sensitivity experiments. For example, *Hirst et al.* [1996] found *less* surface warming in the Southern Ocean using the GM mixing parameterization in the CSIRO coupled model. They attribute this to the confounding effects of differing control climates of the two model versions, especially differences in Antarctic sea-ice extent. *Wiebe and Weaver* [1999] found the expected symmetric warming response when the GM parameterization was used in a simplified coupled model, but differences in Antarctic sea-ice extent also confounded the interpretation of their results somewhat.

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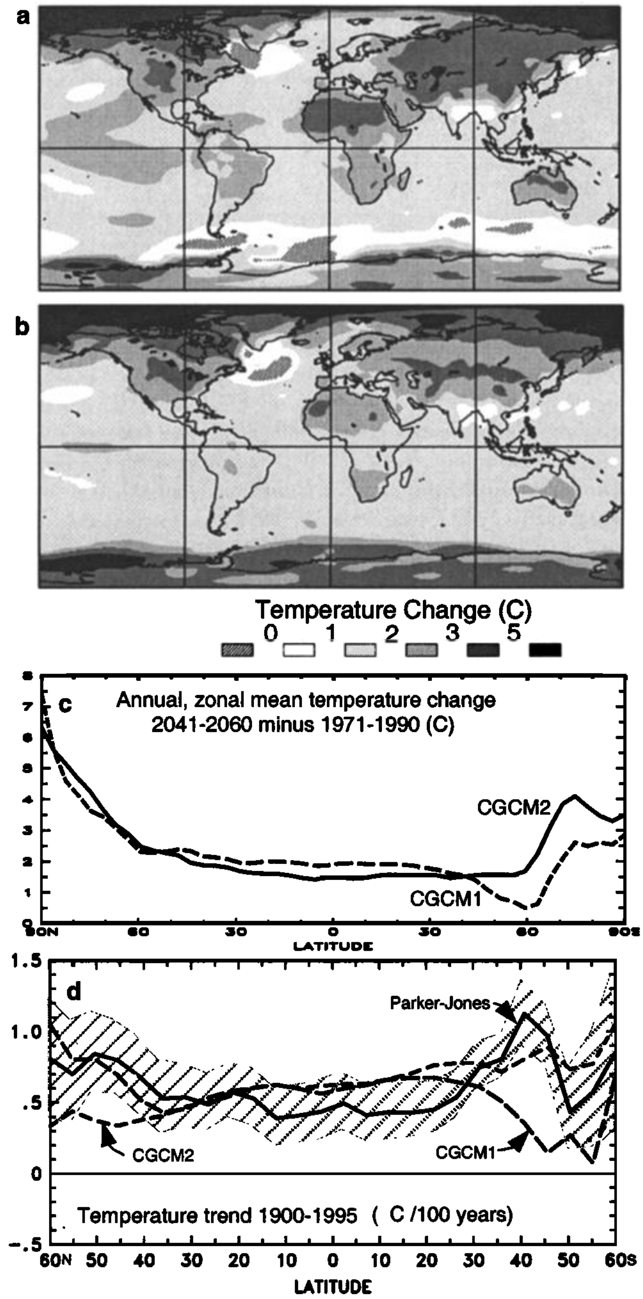

Figure 1. Comparison of CGCM1 and CGCM2 climate warming. (a) Annual mean surface air temperature change from 1971-1990 to 2041-2060, obtained with CGCM1. (b) As in a but for CGCM2. (c) zonal mean of parts a and b. (d) linear trend in zonal mean surface air temperature from 1900 to 1990 compared to observations.

Figure 1d compares simulated and observed trends in zonally-averaged temperature for the period 1900 to 1995. The shaded band around the observationally-based curve is an estimate of the 95% confidence interval (see BFR for a description of the calculation which makes use of the data sets of *Jones* [1994] and *Parker et al.* [1995]). For CGCM1, modeled trends are not distinguishable from those observed in the tropics and at high northern latitudes (data limitations restrict results to latitudes less than 60°), but clearly differ from those observed in the southern extratropics. By contrast, the CGCM2 result is in much better agreement in the SH (although slightly worse in the northern extratropics).

Figure 2 shows that the difference in surface warming accompanying the change in mixing scheme is associated with a difference in heat uptake by the ocean. The ocean warming produced by CGCM1 is characteristic of deep convection in the Southern Ocean which, in CGCM2, is largely confined to the upper 1000m; a result of changes in the depth and location of heat exchange.

The expectation, based on the ocean-only calculation of *Robitaille and Weaver* [1995] and *McDougall et al.* [1996], that global mean surface warming will be larger with the GM versus the HV mixing parameterization, is not evident in our model results. In fact, the change in global, annual mean temperature between 1971-1990 and 2041-2060, and hence the effective climate sensitivity, is nearly identical at 1.96°C for CGCM1 and 1.92°C for CGCM2. Figure 3 compares the

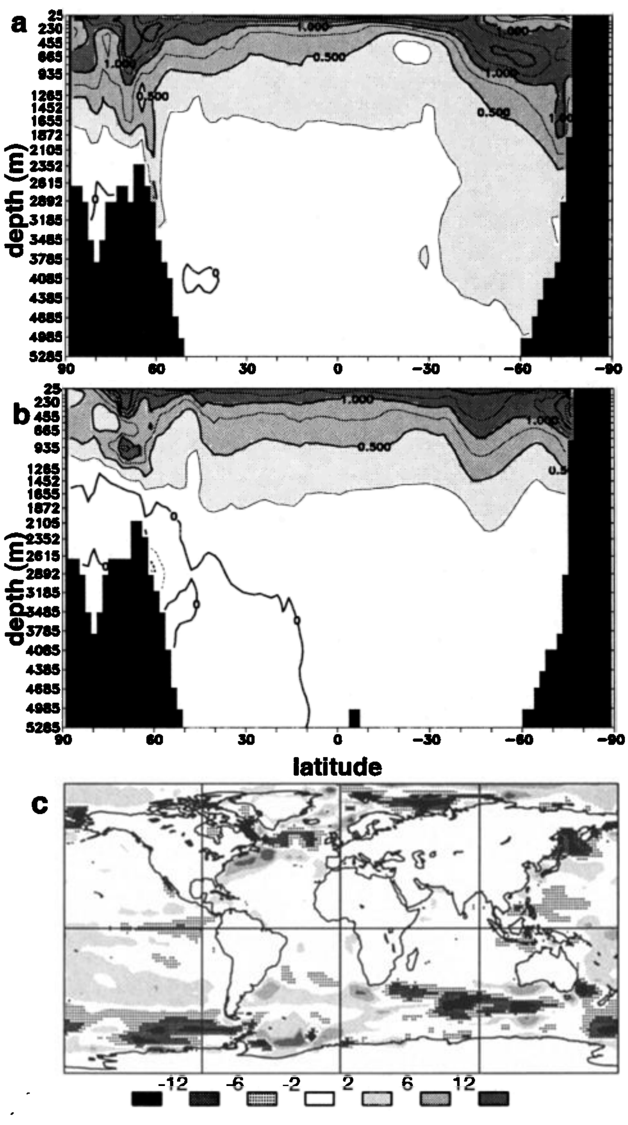

Figure 2. Zonally-averaged ocean temperature change from 1971-1990 to 2041-2060 for: (a) CGCM1; (b) CGCM2. Contour interval is 0.25°C. (c) Difference in net downward heat flux into the ocean, CGCM2 minus CGCM1, averaged over the period 1980-2050.

modeled global sea-level rise due to ocean density changes (the corresponding control run value is subtracted from each to eliminate the influence of climate drift). The differences are small indicating that the change in mixing parameterization affects the geographical location of ocean heat uptake (and, by extension, the uptake of carbon and other atmospheric constituents) but not the globally-averaged value. *Weaver and Wiebe* [1999] and *Jackett et al.* [2000] found slightly larger sea-level differences in more controlled GM versus HV experiments.

Sea-ice extent observations provide an alternative to sparse temperature data in assessing the pattern of modeled climate change. *Cavalieri et al.* [1997] analyze satellite-derived monthly ice extent and find a statistically significant decreasing trend of 2.8% per decade in the NH over the period 1978-1996. By contrast, the trend in SH ice extent is not significantly different from zero, and they interpret this as support for asymmetric warming. Figure 4 displays the annual mean sea-ice extent simulated by CGCM1 and CGCM2. Despite differences in initial coverage associated with differences in control climates, both model versions produce long-term decreasing trends in NH ice extent in reasonable agreement with the observed trend, although only CGCM2's trend is statistically significant over the 1978-1996 period. In the SH, CGCM2 warms more than CGCM1 and so its sea-ice cover exhibits an earlier and more rapid decline. Nevertheless, neither model version exhibits a statistically significant trend in SH ice extent during the 1978-1996 period. For CGCM2, this is because the long-term trend is masked by enhanced variability (likely associated with of sea-ice dynamics). The implication is that the apparent asymmetry in recently-observed hemispheric sea-ice trends is not yet a clear indication of asymmetry in climate warming.

3. Summary and Discussion

Coupled model simulations of climate warming typically show a marked asymmetry with northern high latitudes warming more quickly than southern high latitudes, which warm slowly if at all. The warming simulated with CGCM1, the first version of the CCCma coupled climate model, displays just this asymmetry which is a reflection of the Southern Ocean's ability to sequester heat. By contrast, the results from CGCM2, a more recent model version in which the usual horizontal/vertical (HV) diffusive parameterization of sub-grid ocean mixing is replaced by the Gent-McWillaims (GM) isopycnal eddy stirring parameterization, show a more symmetric warming pattern with reduced heat penetration into the Southern Ocean. The global mean warming and ocean heat uptake (and hence the global mean steric component of sea-level rise) nevertheless remain basically the same in both model versions.

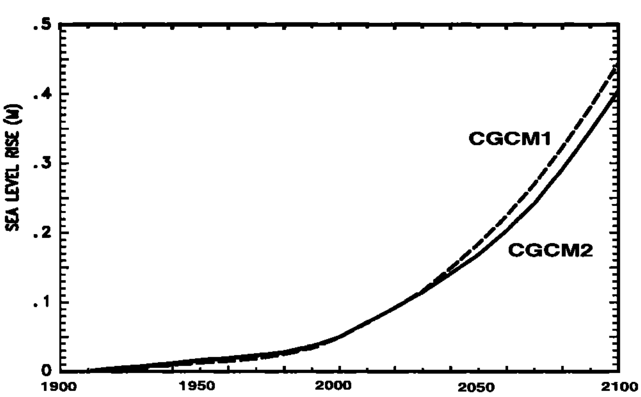

Figure 3. Global, decadal mean steric sea-level rise relative to the corresponding control simulation.

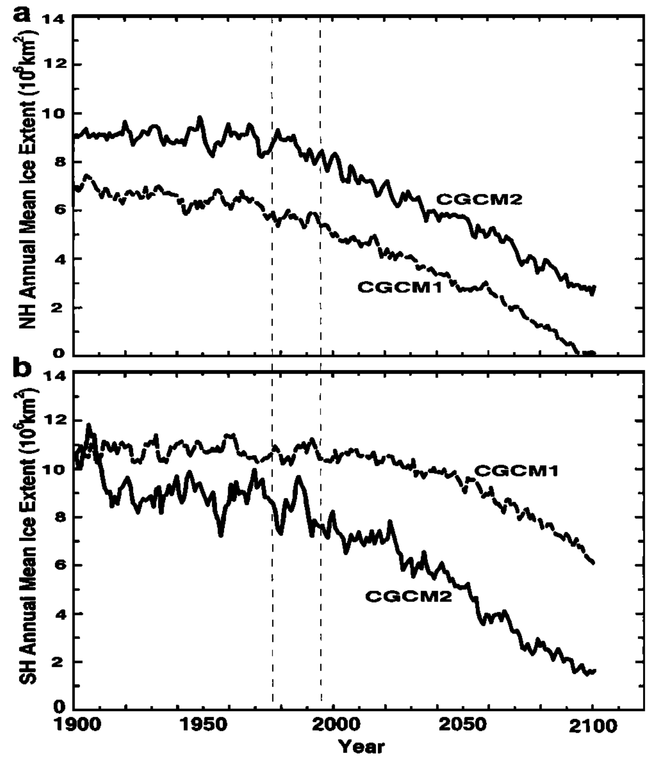

Figure 4. Simulated annual mean sea-ice extent: (a) northern hemisphere; (b) southern hemisphere. The vertical dashed lines indicate the 1978 to 1996 period of satellite observations referred to in the text.

Although not designed as a sensitivity experiment, these results can be qualitatively compared to previous ocean-only and simplified coupled model experiments isolating the effect of HV versus GM mixing. These suggested a more symmetric warming response for the GM parameterization, in agreement with the results found here. The earlier experiments also suggested a larger global average warming, whereas the two model versions compared here exhibit almost identical warming.

The more rapid SH warming produced by CGCM2 is in somewhat better agreement with observed temperature trends for the 20th century. Both model versions simulate decreasing trends in NH ice extent which agree reasonably well with the observed trend from 1978-1996. Neither model version produces a significant trend in SH ice extent over the same period, again in agreement with observations, despite the substantially greater warming in this region simulated by CGCM2. The short satellite record of sea-ice extent is apparently not (yet) able to distinguish between the patterns of warming simulated by the two versions of the model.

The CCCma coupled model results imply that climate change simulations are sensitive to the parameterization of ocean mixing processes, with the CGCM2 version producing greater surface warming in the Southern Ocean and less hemispheric asymmetry compared to results from other current-generation coupled models. Since CGCM2 employs physical parameterizations which are more plausible than those of its predecessor and produces 20th-century warming trends which are in modestly better accord with observations, the more rapid southern hemisphere warming simulated by this version of the model should be considered a realistic possibility.

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